Data Communication and Computer Networks

Chapter 3 | Data Link Layer

Unit 3 - Error control, data link protocols, multiple access, Ethernet and WLAN (8 hrs)

Syllabus covered

Error detection and correction - types of errors, Hamming distance, parity, Hamming code, CRC, checksum; data link control - framing, flow and error control, stop-and-wait ARQ, Go-Back-N ARQ, selective repeat ARQ, HDLC, PPP; multiple access - random access, controlled access, channelization; LAN - standard Ethernet, IEEE 802.11 WLAN.

What is inside

Section	Question type	Marks	Count
Section A	Short answer - definitions, one-liners, small numericals	2	44
Section B	Long answer - full explanations with tables and diagrams	7	14

How to use this bank

- Section A is dominated by small numericals: Hamming distance, burst-error length, CRC and checksum. Practise the four worked examples until they are automatic.
- For the ARQ questions memorise the window sizes: stop-and-wait 1, Go-Back-N 2^m - 1 with a receive window of 1, selective repeat at most half of 2^m.
- Draw every protocol answer as a time-line with sender on the left and receiver on the right.
- Random access, controlled access and channelization are three separate families - never mix them in one answer.

Source tags

- **PYQ <year>** - asked in a C. V. Raman Global University end-semester paper (2022, 2023, 2024-25 or the 2024-25 backlog paper).
- **Model** - from the printed unit-wise model question paper.
- **Assignment** - from the course assignment for this unit.
- **Notes** - from the lecture slides, not yet asked but examinable.

Section A | 2-mark questions

Answer in two to four lines. Each point below is one mark-bearing idea, so a single line written in the exam earns the mark.

A1. Define an error and name the two types. [Notes]

- An error means the message received is **not identical** to the message sent.
- **Single-bit error** - only one bit of the data unit changes, 1 to 0 or 0 to 1.
- **Burst error** - two or more bits change, and the burst length is measured from the first corrupted bit to the last.

A2. State burst error. If the sent pattern is 1100010001000010 and the received pattern is 1101110111000010, find the length of the burst error. [PYQ 2022]

1. XOR the two patterns: 0001100110000000.
2. The first corrupted bit is the **4th** and the last is the **9th**.
3. Burst length = 9 - 4 + 1 = **6 bits**.
4. Note: the burst length counts from the first to the last error, not the number of corrupted bits.

A3. What is Hamming distance? [Notes]

- The Hamming distance d(x, y) between two words is the **number of corresponding bits that differ** - it is found by XOR-ing them and counting the 1s.
- The **minimum Hamming distance** dmin of a code is the smallest distance between any pair of valid codewords.
- dmin is the single measure of how strong a code is.

A4. Find the minimum Hamming distance for the given pairs: (00111, 01011) and (01011, 10101). [PYQ 2024-25]

1. d(00111, 01011): the 2nd and 3rd bits differ, so d = **2**.
2. d(01011, 10101): bits 1, 2, 3 and 4 differ, so d = **4**.
3. dmin for the pair of words = **2**.

A5. Find the Hamming distance for each of the following: d(10000, 00000), d(10101, 10000) and d(00000, 11111). [PYQ 2022]

1. d(10000, 00000) = **1** - only the first bit differs.
2. d(10101, 10000) = **3** - bits 3, 4 and 5 differ.
3. d(00000, 11111) = **5** - every bit differs.

A6. Find the minimum Hamming distance between the pairs (00000, 01011) and (01011, 10101). [PYQ 2023]

1. d(00000, 01011) = **3** - bits 2, 4 and 5 differ.
2. d(01011, 10101) = **4** - bits 1, 2, 3 and 4 differ.

A7. What minimum Hamming distance is needed to detect s errors and to correct t errors? [Notes]

- To **detect s errors**: dmin = s + 1, because s errors cannot turn one valid codeword into another.
- To **correct t errors**: dmin = 2t + 1, because the received word must stay closer to the original than to any other codeword.
- Example: dmin = 3 detects 2 errors and corrects 1.

A8. Explain the single parity check code and state what it can detect. [Notes]

- One redundant bit is appended so that the total number of 1s is even (even parity) or odd (odd parity).
- It is a linear block code with n = k + 1 and dmin = 2.
- It detects **any odd number** of errors but cannot correct any, and it misses every even number of errors.

A9. What is a linear block code? [Notes]

- A block code maps k data bits to n = k + r codeword bits using a substitution table.
- It is **linear** if the modulo-2 sum (XOR) of any two valid codewords is also a valid codeword.
- Examples: single parity check, Hamming, cyclic (CRC), Reed-Solomon, BCH, Golay.

A10. How many parity bits are needed in a Hamming code? [Notes]

- r parity bits are placed at positions 1, 2, 4, 8 and so on; data bits fill the rest.
- Condition: **2^r is greater than or equal to m + r + 1**, where m is the number of data bits.
- For m = 4, r = 3 satisfies it: 2^3 = 8 and m + r + 1 = 4 + 3 + 1 = 8. That is the (7,4) Hamming code.

A11. For transmission of the data word 1011, find the transmitted codeword using the (7,4) Hamming code with even parity. [Assignment 3; PYQ 2023]

1. Place the data in positions 7, 6, 5, 3: d7 = 1, d6 = 0, d5 = 1, d3 = 1.
2. P1 (position 1) = d3 XOR d5 XOR d7 = 1 XOR 1 XOR 1 = **1**.
3. P2 (position 2) = d3 XOR d6 XOR d7 = 1 XOR 0 XOR 1 = **0**.
4. P4 (position 4) = d5 XOR d6 XOR d7 = 1 XOR 0 XOR 1 = **0**.
5. Codeword, position 7 down to 1: **1 0 1 0 1 0 1**.
6. Check: positions (1,3,5,7) hold 1,1,1,1 - four 1s, even. Positions (2,3,6,7) hold 0,1,0,1 - two 1s. Positions (4,5,6,7) hold 0,1,0,1 - two 1s.

A12. What is CRC and what is its purpose? [PYQ 2022; PYQ 2024-25]

- Cyclic redundancy check is a **cyclic code** used for error detection: the dataword is extended by n-k zeros and divided modulo-2 by an agreed generator polynomial.
- The **remainder** is the CRC, and it replaces the appended zeros to form the codeword.
- At the receiver the whole codeword is divided again; a **non-zero remainder** means an error.
- It is very strong against burst errors and cheap in hardware (shift registers), which is why Ethernet and HDLC use it.

A13. Given the data word 1010001101 and the divisor 110101, find the CRC. [PYQ 2024-25]

1. The divisor has 6 bits, so append 5 zeros: 1010001101**00000**.
2. Divide modulo-2 (XOR) by 110101, taking 6 bits at a time.
3. Remainder = **01110**, so CRC = 01110.
4. Codeword sent = **1010001101 01110**.
5. Verification: dividing the codeword by 110101 gives remainder **00000**, which proves it is error-free.

A14. Given the 9-bit sequence 101100101 and the divisor polynomial x^4 + x^2 + 1, find the CRC code. [PYQ 2022]

1. x^4 + x^2 + 1 = **10101** (5 bits), so append 4 zeros: 101100101**0000**.
2. Divide modulo-2 by 10101.
3. Remainder = **1000**.
4. Codeword = **101100101 1000**; dividing it again by 10101 leaves 0000.

A15. Given the data word 101001111 and the divisor 10111, show the generation of the CRC codeword. [Assignment 3]

1. Divisor has 5 bits, so append 4 zeros: 101001111**0000**.
2. Divide modulo-2 by 10111.
3. Remainder = **0101**.
4. Codeword sent = **101001111 0101**; the receiver's division leaves 0000, verifying no error.

A16. What is a cyclic code? [Notes]

- A cyclic code is a linear block code with one extra property: any **cyclic shift** of a codeword is also a valid codeword.
- Example: 1011000 rotated left becomes 0110001, still valid.
- The property allows the code to be built with shift registers, so CRC encoding and checking are cheap.

A17. What is checksum? How does it work? [Notes]

- An error-detecting technique for a message of any length, used mostly at the network and transport layers.
- The message is divided into m-bit units; the sender adds them and sends the **complement of the sum** as the checksum.
- The receiver adds all units including the checksum; if the result is all 0s the message is accepted, otherwise it is discarded.

A18. A sender has two data items to send: (4567)16 and (B498)16. Find the checksum. [PYQ 2024-25 backlog]

1. Add: 4567 + B498 = **F9FF**.
2. The checksum is the one's complement of the sum: FFFF - F9FF = **0600**.
3. Check: F9FF + 0600 = FFFF, that is all 1s - so the receiver's addition gives zero after complementing.

A19. Describe the need for framing. [PYQ 2024-25 backlog]

- The physical layer delivers only a **stream of bits**, with no boundaries; the data link layer must group them into **frames** so that each frame is distinguishable from the next.
- Framing is what makes addressing, error control and flow control possible - you cannot check or acknowledge a frame you cannot identify.
- Methods: character-oriented (byte stuffing) and bit-oriented (bit stuffing).

A20. Describe bit stuffing and its requirement in a bit-oriented protocol. [PYQ 2022]

- The flag 01111110 marks the start and end of a frame.
- If the data contain five consecutive 1s, the sender inserts a **0** after them, so the pattern 01111110 can never occur inside the data.
- The receiver removes the extra 0 after five 1s - that is **unstuffing**.
- **Requirement**: without stuffing, data that happen to contain the flag would be mistaken for the end of the frame.

A21. What is byte stuffing? [Notes]

- A character-oriented framing method: a flag byte marks the frame boundary.
- If the data contain the flag byte or the escape byte, the sender inserts an **extra escape byte** before it.
- The receiver removes the escape byte - unstuffing.

A22. Define flow control and error control. [PYQ 2024-25 backlog]

- **Flow control** - the set of procedures that restrict how much data the sender may send before waiting for an acknowledgement, so a fast sender cannot drown a slow receiver.
- **Error control** - based on automatic repeat request (ARQ): the sender keeps a copy, a timer runs, and a frame is retransmitted when it is lost or corrupted.
- Together they are called **data link control**.

A23. Describe the mechanism of stop-and-wait ARQ. [PYQ 2022; PYQ 2024-25 backlog]

- The sender sends **one frame**, starts a timer and waits.
- The receiver replies with an ACK that announces the number of the **next frame expected**.
- If the ACK arrives, the sender sends the next frame; if the timer expires, it retransmits the same frame.
- Sequence numbers are **modulo 2** (0, 1, 0, 1), because only two frames are ever outstanding.
- Weakness: on a long, fast link the sender idles most of the time, so utilisation is poor.

A24. Give the window sizes used in Go-Back-N and selective repeat ARQ. [Notes]

- Sequence numbers are modulo **2^m**, where m is the size of the sequence-number field.
- **Go-Back-N**: send window size 2^m - 1; receive window size **1**.
- **Selective repeat**: send and receive windows each at most **half** of 2^m.
- The receive window of 1 is exactly why GBN throws away out-of-order frames.

A25. Differentiate Go-Back-N ARQ and selective repeat ARQ. [PYQ 2024-25 backlog; PYQ 2022]

Point	Go-Back-N	Selective repeat
Window size	Send 2^m - 1, receive 1	Both at most half of 2^m
On a lost frame	Discard all later frames, resend N frames	Store later frames, resend only the lost one
ACK	Cumulative	Individual, plus NAK
Timers	One timer for the oldest frame	One timer per outstanding frame
Efficiency	Lower on error-prone links	Higher, but more complex
Buffering	Not needed at the receiver	Required at the receiver

A26. State the advantages of Go-Back-N ARQ over stop-and-wait ARQ. [PYQ 2024-25]

- It keeps up to **2^m - 1 frames in flight**, so the link is not left idle while an ACK travels - utilisation rises sharply.
- **Cumulative acknowledgements** mean one ACK can cover several frames, so a lost ACK is often harmless.
- One timer serves the whole window instead of one per frame.
- The receiver needs no buffer, which keeps it simple.

A27. What is HDLC? Draw the HDLC frame formats and state the use of each. [PYQ 2024-25; PYQ 2023]

- High-level Data Link Control is a **bit-oriented** protocol for point-to-point and multipoint links that implements the ARQ mechanisms.
- Frame: **flag, address, control, information, FCS, flag**; the flag is 01111110 and FCS is a 2- or 4-byte CRC.
- **I-frame** - carries user data plus control information relating to it; sequence and piggybacked ACK numbers.
- **S-frame** - carries only control information: receive-ready, reject, receive-not-ready.
- **U-frame** - system management: link setup, disconnect, mode setting.

A28. Name the transfer modes of HDLC. [Notes]

- **NRM** (normal response mode) - unbalanced: one primary and many secondaries; a secondary may transmit only when polled. Used on multipoint links.
- **ARM** (asynchronous response mode) - unbalanced, but a secondary may transmit without permission.
- **ABM** (asynchronous balanced mode) - balanced: point-to-point, each station can act as primary or secondary.

A29. What is PPP and which states does it pass through? [Notes]

- Point-to-Point Protocol is a **byte-oriented** protocol for point-to-point access, the standard for dial-up and ISP links.
- States: **dead** (no carrier), **establish** (options negotiated, optional authentication), **authenticate**, **open** (data transfer), **terminate** (carrier dropped, back to dead).
- It carries any network-layer protocol in its payload and uses byte stuffing with a flag byte.

A30. What is multiple access and why is it needed? [PYQ 2022; PYQ 2024-25 backlog]

- When many stations share **one common link**, rules are needed to decide who may transmit and when.
- Without them, simultaneous transmissions **collide** and both frames are destroyed.
- These rules form a sublayer of the data link layer called **media access control** (MAC).
- Three families: random access, controlled access and channelization.

A31. Differentiate a random access protocol from a controlled access method. [PYQ 2024-25]

Point	Random access	Controlled access
Who decides	The station itself decides to send	Other stations authorise it
Hierarchy	None - all stations equal	A primary, or a token, or reservations
Collision	Possible, and must be handled	Avoided by design
Efficiency	Good at low load, poor at high load	Predictable, better at high load
Examples	ALOHA, CSMA, CSMA/CD, CSMA/CA	Reservation, polling, token passing

A32. What is the maximum throughput of pure ALOHA? Give the derivation. [PYQ 2024-25 backlog]

1. Let G be the offered load in frames per frame-time, and S the throughput.
2. A frame is undamaged only if **no other frame starts within one frame time before or after** it - a vulnerable period of 2 frame times.
3. Probability of no collision = e^-2G, so **S = G e^-2G**.
4. Differentiate and set to zero: maximum at G = 0.5, giving S = 1/(2e) = **0.184, that is 18.4 per cent**.

A33. What is the maximum throughput of slotted ALOHA? [Notes]

- In slotted ALOHA stations may transmit only at slot boundaries, so the vulnerable period halves to one frame time.
- **S = G e^-G**, maximum at G = 1.
- S = 1/e = **0.368, that is 36.8 per cent** - double that of pure ALOHA, at the cost of synchronisation.

A34. Explain the CSMA persistence methods. [PYQ 2024-25 backlog]

- **1-persistent** - sense the channel; if idle, transmit immediately with probability 1. Simple, but collisions are certain when two stations find the line idle together.
- **Non-persistent** - if the line is busy, wait a **random** time and sense again. Reduces collisions, but the channel can stay idle while someone is backing off.
- **p-persistent** - when the line is idle, transmit with probability p and wait one slot with probability q = 1 - p; if the slot is then idle, repeat. It combines the best of both.

A35. Explain the operation of CSMA/CD. [PYQ 2022; PYQ 2024-25]

- **Sense** the medium before transmitting; if busy, wait.
- If idle, start transmitting and **keep sensing while transmitting**.
- **Collision detected**: send a jam signal, abort the frame, and wait a random **backoff** time (binary exponential).
- Retransmit after the backoff. It is used in traditional Ethernet.
- Requirement: the frame must be long enough that the sender is still transmitting when the collision signal returns, that is transmission time is at least twice the propagation delay.

A36. Explain CSMA/CA and its three strategies. [PYQ 2022]

- In wireless LANs a station cannot hear collisions, so it must **avoid** them instead of detecting them.
- **Interframe space** - an idle station waits an IFS before sending; the length of the IFS also gives priority (SIFS for an ACK, longer for data).
- **Contention window** - the station picks a random number of slots to wait, so two stations rarely pick the same moment.
- **Acknowledgement** - the receiver ACKs every frame; absence of an ACK means the frame was lost and it is retransmitted.

A37. Name the three controlled access methods. [Notes]

- **Reservation** - a reservation frame with N mini slots precedes the data; a station reserves its own mini slot.
- **Polling** - a primary station polls the secondaries in round-robin order with a select or poll frame; a primary failure stops the network.
- **Token passing** - a token circulates round a logical ring; a station may send only when it holds the token.

A38. What is channelization? Name its three protocols. [PYQ 2024-25 backlog]

- Channelization is a multiple-access method in which the available bandwidth of a link is shared **in frequency, in time or by code**.
- **FDMA** - the band is divided into frequency channels separated by guard bands.
- **TDMA** - one channel, time-shared in slots.
- **CDMA** - one channel carries all transmissions at once, separated by orthogonal codes (Walsh codes).

A39. Explain the difference between FDMA and CDMA. [PYQ 2024-25 backlog]

Point	FDMA	CDMA
Shared resource	Frequency band	Code
Separation	Guard bands between channels	Orthogonal Walsh codes
Simultaneous users	Yes, on separate frequencies	Yes, on the same frequency
Bandwidth per user	Fixed slice	The whole band
Effect of noise	Confined to one channel	Spread, so it is averaged out

A40. Briefly explain the types of network configurations provided by IEEE 802.11. [PYQ 2024-25]

- **BSS** (basic service set) - the building block: wireless stations plus an optional access point.
- **Ad hoc network** - a BSS **without** an AP; stations talk directly to each other.
- **Infrastructure network** - a BSS **with** an AP; all traffic goes through the AP.
- **ESS** (extended service set) - two or more BSSs with APs joined by a **distribution system**, which is what lets a user roam across access points.

A41. State the two types of services provided by IEEE 802.11. [PYQ 2022]

- **Station services** - **authentication** (prove identity before association), **deauthentication** (end the association), and **privacy** (encryption of the wireless traffic).
- **Distribution system services** - **association** (a station joins an AP), **disassociation**, **reassociation** (moving to another AP), and **distribution** (routing frames through the DS).
- Some texts group these as BSS services and ESS services.

A42. List the fields of the IEEE 802.3 MAC frame. [Notes]

- **Preamble** - 7 bytes of alternating 1s and 0s: alert and timing pulse.
- **SFD** - 10101011, the last chance to synchronise.
- **Destination address** - 6 bytes; the least significant bit of the first byte decides unicast (0) or multicast (1).
- **Source address** - 6 bytes.
- **Length or type** - 2 bytes.
- **Data** - 46 to 1500 bytes, plus padding.
- **CRC** - 4 bytes for error detection.

A43. Differentiate thicknet and thin Ethernet. [PYQ 2023]

Point	Thick Ethernet (10Base5)	Thin Ethernet (10Base2)
Cable	Thick coaxial, 10 mm	Thin coaxial, 5 mm
Segment length	500 m per segment	185 m per segment
Transceiver	External transceiver with a vampire tap	Transceiver built into the NIC
Cost and handling	Expensive, stiff, hard to install	Cheaper, flexible, easy to install
Stations per segment	Up to 100	Up to 30

A44. A LAN runs at 8000 Mbps with 5000-bit frames and a signal speed of 4,000,000 m/s. Find the maximum cable length for CSMA/CD to work. [PYQ 2023]

1. Transmission time of one frame: Tfr = 5000 / 8 x 10^9 = 0.625 microseconds.
2. For collision detection the sender must still be transmitting when the collision signal returns: 2 x Tprop is at most Tfr.
3. So Tprop is at most 0.3125 microseconds.
4. L = speed x time = 4 x 10^6 x 0.3125 x 10^-6 = **1250 m**.

Section B | 7-mark questions

Write the heading points first, then expand each one. The bullet structure below is the marking scheme: one bullet is worth roughly half a mark to a mark.

B1. Explain cyclic redundancy check. Given the data word 1010001101 and the divisor 110101, find the CRC and verify your answer. [PYQ 2024-25; PYQ 2022; Assignment 3]

Principle. CRC is a cyclic code. The dataword is augmented with n-k zeros and divided modulo-2 by a generator polynomial agreed between sender and receiver; the remainder is the CRC and is appended to the data to form the codeword. The receiver divides the whole codeword by the same generator - a zero remainder means no error.

1. Divisor 110101 has 6 bits, so append **5 zeros**: 1010001101 00000.
2. Take the first 6 bits, 101000, and XOR with 110101, giving 011101; bring down the next bit and repeat.
3. Continue the division to the end of the augmented word.
4. **Remainder = 01110**, so the CRC is 01110 and the transmitted codeword is **1010001101 01110**.

Verification. Divide the received codeword 101000110101110 by 110101. The remainder is **00000**, so the receiver accepts the frame. If any bit had been corrupted the remainder would be non-zero, and the frame would be discarded and retransmitted.

Memory hook - three steps only: **append zeros, divide modulo-2, put the remainder in place of the zeros**. The same division at the receiver is the check.

B2. Explain the Hamming code. For the data word 1011, construct the (7,4) codeword with even parity and show how a single-bit error is located and corrected. [Assignment 3; PYQ 2023]

Principle. Hamming code is a linear block code that adds r parity bits to m data bits so that n = m + r. Parity bits occupy the positions that are powers of 2 (1, 2, 4, 8 ...), each parity bit checks the positions whose index contains its bit, and the syndrome formed from the failing checks gives the **position** of the error.

1. m = 4, r = 3 satisfies 2^r is greater than or equal to m + r + 1, so the code is (7,4).
2. Data placed at positions 7, 6, 5, 3: d7 = 1, d6 = 0, d5 = 1, d3 = 1.
3. P1 = d3 XOR d5 XOR d7 = 1 XOR 1 XOR 1 = **1**.
4. P2 = d3 XOR d6 XOR d7 = 1 XOR 0 XOR 1 = **0**.
5. P4 = d5 XOR d6 XOR d7 = 1 XOR 0 XOR 1 = **0**.
6. **Codeword = 1010101** (positions 7 to 1).
7. Suppose bit 4 flips: received = 1011101. Syndrome S1 = positions 1,3,5,7 = 1,1,1,1 = 0; S2 = positions 2,3,6,7 = 0,1,0,1 = 0; S4 = positions 4,5,6,7 = 1,1,0,1 = 1.
8. Syndrome S4 S2 S1 = **100**, that is position 4 - exactly the corrupted bit. Flip it back and the data 1011 is recovered.

Memory hook - the syndrome *is* the error position in binary. That is the whole elegance of Hamming codes.

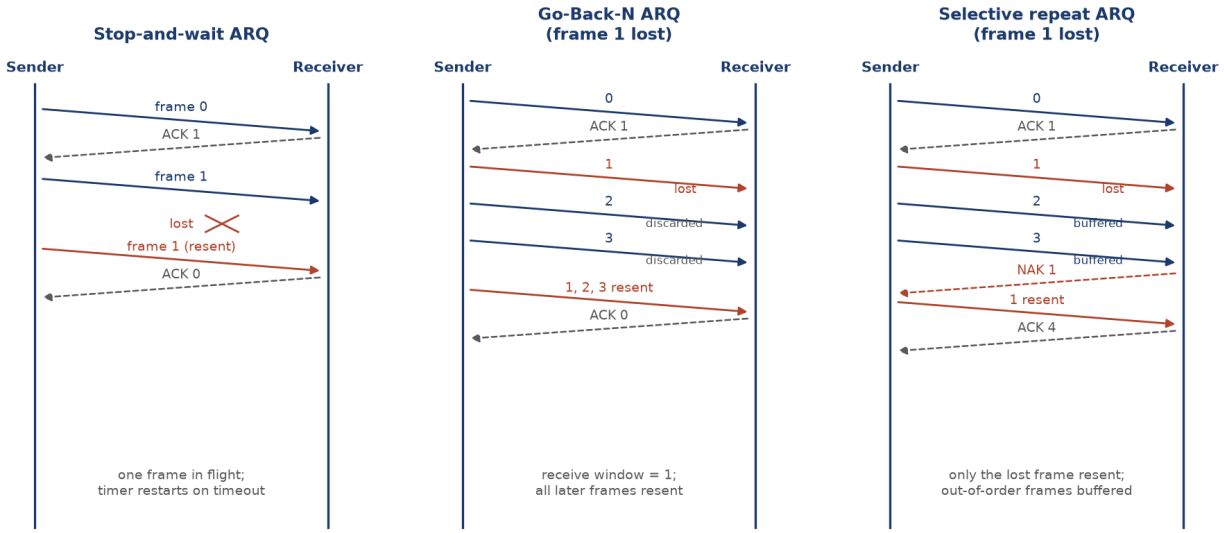
B3. Compare the error detection and correction techniques used at the data link layer. [Notes]

Technique	Redundancy	Detects	Corrects	Use
Single parity	1 bit	Odd number of errors	No	Simple links, character codes
Two-dimensional parity	Row and column bits	Most bursts	No	Older protocols
Hamming code	r bits, 2^r >= m+r+1	2-bit errors	1-bit error	Memory, satellite links
CRC	n-k bits (16 or 32)	All single, double and odd errors, most bursts	No	Ethernet, HDLC, Wi-Fi
Checksum	m bits	Most errors	No	IP, TCP, UDP headers

- **Detection** needs dmin = s + 1; **correction** needs dmin = 2t + 1.
- CRC gives the strongest detection for the least hardware, which is why the data link layer uses it; checksum is used higher up because it is cheap in software.

B4. Describe the mechanism of stop-and-wait ARQ and distinguish between Go-Back-N ARQ and selective repeat ARQ. [PYQ 2022; PYQ 2024-25 backlog]

Stop-and-wait ARQ. The sender transmits one frame and starts a timer; the receiver acknowledges with the number of the next frame expected; if the ACK arrives the next frame is sent, if the timer expires the frame is retransmitted. Sequence numbers are modulo 2. It is correct but slow: with a long propagation delay the sender is idle most of the time.



Point	Go-Back-N	Selective repeat
Window	2^m - 1 send, 1 receive	Both at most 2^m / 2
On error	Resend the lost frame and all after it	Resend only the lost frame
Receiver	Discards out-of-order frames	Buffers them until the gap is filled
ACK	Cumulative	Individual ACKs and NAKs
Timer	One for the oldest frame	One per outstanding frame
Complexity and efficiency	Simple, wasteful	Complex, efficient

B5. Explain Go-Back-N ARQ with a diagram and state its advantages over stop-and-wait ARQ. [PYQ 2024-25]

1. Sequence numbers are modulo 2^m; the send window holds up to 2^m - 1 frames, the receive window is 1.
2. The sender may transmit up to the window limit without waiting; each frame carries a sequence number and a timer runs for the oldest outstanding frame.
3. The receiver accepts only the frame with the expected number and returns a **cumulative ACK**; any out-of-order frame is discarded.
4. If the timer of the oldest frame expires, the sender resends that frame **and every frame after it** - hence the name.
5. Diagram: sender on the left, receiver on the right, frames 0, 1, 2, 3 sent; frame 1 is lost, so 2 and 3 are discarded at the receiver and after the timeout all three are resent.

- **Advantages**: many frames in flight, so utilisation is far better than stop-and-wait; one ACK can acknowledge several frames; one timer for the whole window; no buffering at the receiver.

B6. Explain selective repeat ARQ. [Notes]

- Both the send and receive windows are at most **half of 2^m**, so a frame can always be distinguished from a retransmission.
- The receiver **buffers** out-of-order frames and marks them; they are delivered to the network layer only when the missing frame arrives.
- The sender keeps a **timer per frame** and resends only the frame whose timer expires; a NAK can request one frame immediately.
- ACKs are sent when data are delivered to the network layer, so one ACK may cover several frames.
- Cost: buffering and per-frame timers - more complexity for much better performance on error-prone links.

B7. Write short notes on (a) flow and error control and (b) stop-and-wait ARQ. [PYQ 2024-25 backlog]

- **Flow control** - procedures that limit how much the sender may send before an acknowledgement, so the receiver's buffer is not overrun.
- **Error control** - ARQ: the sender keeps a copy of every frame, a timer detects loss, an ACK or NAK detects success or failure, and the frame is retransmitted when needed. Sequence numbers stop the receiver accepting duplicates.
- **Stop-and-wait ARQ** - one frame at a time with modulo-2 sequence numbers and a timeout. Simple and correct, but the sender idles while waiting for each ACK, so the utilisation of a long fast link is very low; the sliding-window protocols exist to fix exactly that.

B8. Explain HDLC: the services it provides and its frame structure. [PYQ 2023]

- **HDLC** is a bit-oriented data link protocol for point-to-point and multipoint links, providing framing, flow control and error control through ARQ.
- **Frame format**: flag (8 bits, 01111110) - address (8 or more) - control (8 or 16) - information (variable) - FCS (16 or 32) - flag.
- **Flag** delimits the frame and synchronises the receiver; bit stuffing prevents it appearing inside the data.
- **Address** identifies the secondary station; **control** carries sequence numbers, ACK and frame type; **FCS** is the CRC.
- **Three frame types**: I-frames carry user data with piggybacked ACK; S-frames carry control only (receive-ready, reject, receive-not-ready); U-frames manage the link (set mode, disconnect).
- **Transfer modes**: NRM (unbalanced, secondary polled), ARM (unbalanced, secondary may initiate) and ABM (balanced, point-to-point).

Compare with PPP: HDLC is bit-oriented and can also run multipoint; PPP is byte-oriented, point-to-point only, and adds authentication and protocol multiplexing.

B9. Explain framing and the two stuffing techniques. [PYQ 2022]

- **Framing** packs the bit stream into frames so each frame can be identified, addressed, checked and acknowledged.
- **Character-oriented framing** uses a flag byte; if the flag or the escape byte occurs in the data, the sender inserts an extra escape byte (**byte stuffing**), which the receiver removes.
- **Bit-oriented framing** uses the flag 01111110; the sender inserts a 0 after every five consecutive 1s (**bit stuffing**) so the flag can never occur in the data.
- The receiver counts five 1s and removes the following 0 - unstuffing.
- Bit stuffing is transparent to any data pattern and is used by HDLC; byte stuffing is simpler but ties the frame to a byte size.

B10. Differentiate a random access protocol from a controlled access method, and explain the operation of CSMA/CD. [PYQ 2024-25; PYQ 2022]

Point	Random access	Controlled access
Decision	Station decides for itself	Stations consult, or a primary authorises
Collisions	Possible; must be detected or avoided	Designed out
Behaviour at high load	Degrades - more collisions	Stable

Point	Random access	Controlled access
Examples	ALOHA, CSMA, CSMA/CD, CSMA/CA	Reservation, polling, token passing

1. CSMA/CD: sense the carrier. If the medium is busy, wait; if idle, transmit.
2. Continue sensing **while** transmitting - Ethernet uses a simultaneous transmit and receive path for this.
3. On detecting a collision, send a 32-bit jam signal so all stations know, abort the frame and increment the backoff counter.
4. Wait a random time chosen from 0 to 2^k - 1 slot times (binary exponential backoff) and try again.
5. The frame must be at least 64 bytes so that a collision at the far end still arrives before the sender finishes - which is why the maximum segment length is tied to the minimum frame size.

B11. Explain CSMA with its persistence methods and compare CSMA/CD with CSMA/CA. [PYQ 2024-25 backlog; PYQ 2022]

- **1-persistent** - transmit immediately when idle; greedy, high collision rate.
- **Non-persistent** - if busy, back off a random time and sense again; fewer collisions but the channel can idle.
- **p-persistent** - when idle, transmit with probability p and defer one slot with probability 1 - p; a compromise for slotted channels.

Point	CSMA/CD	CSMA/CA
Detects collision	Yes, during transmission	No - cannot hear while sending
Strategy	Detect and recover	Avoid before it happens
Mechanisms	Sensing plus jam signal and backoff	Interframe space, contention window, ACKs
Medium	Wired Ethernet	Wireless LAN (802.11)
Efficiency	High on a wired bus	Lower, because of ACK overhead

B12. Explain pure and slotted ALOHA and derive their maximum throughput. [PYQ 2024-25 backlog]

Pure ALOHA. A station transmits whenever it has a frame, without sensing. Two frames collide if they overlap at all.

1. A frame survives only if no other frame begins within one frame time before or after it, so the vulnerable period is **2 frame times**.
2. With Poisson arrivals of offered load G, the probability of no other arrival in 2 frame times is e^-2G.
3. **S = G e^-2G**; differentiating gives the maximum at G = 0.5, so Smax = 1/(2e) = **0.184**.
4. **Slotted ALOHA** allows transmission only at slot boundaries, so the vulnerable period falls to one frame time: **S = G e^-G**, maximum at G = 1, Smax = 1/e = **0.368**.
5. So slotted ALOHA doubles the throughput at the cost of synchronising all stations to a common clock.

Remember the two numbers: 18.4 per cent for pure ALOHA, 36.8 per cent for slotted ALOHA - and both fall far below CSMA, which is why ALOHA is only history.

B13. Differentiate among the various methods of multiple access. [PYQ 2024-25 backlog; Assignment 3]

Family	Principle	Members	Behaviour
Random access	Stations contend; collisions handled	ALOHA, CSMA, CSMA/CD, CSMA/CA	Good at light load, poor at heavy load
Controlled access	Stations are authorised to send	Reservation, polling, token passing	Predictable, stable at heavy load
Channelization	Bandwidth pre-divided by frequency, time or code	FDMA, TDMA, CDMA	No contention, but capacity is reserved even when idle

- **Choose by load**: random access for bursty traffic on a lightly loaded LAN; controlled access for deterministic delay, as in industrial networks; channelization for continuous streams such as voice.
- **Cost**: random access wastes capacity on collisions, controlled access wastes it on polling or token overhead, channelization wastes it on idle reserved slots.

B14. Write short notes on (a) standard Ethernet (IEEE 802.3) and (b) wireless LAN (IEEE 802.11). [Assignment 3]

- **IEEE 802.3 Ethernet** - Project 802 began in 1985 to standardise the physical and data link layers of LANs. The MAC frame is preamble, SFD, destination address (6 bytes), source address (6 bytes), length or type, data (46 to 1500 bytes) and CRC (4 bytes). The least significant bit of the first destination byte distinguishes unicast from multicast. Access is CSMA/CD. Categories include 10Base5 (thick coax, 500 m), 10Base2 (thin coax, 185 m), 10Base-T (twisted pair, 100 m) and 10Base-F (fiber).
- **IEEE 802.11 WLAN** - the building block is the basic service set: wireless stations with an optional access point. A BSS without an AP is an ad hoc network; with an AP it is an infrastructure network. Two or more BSSs joined by a distribution system form an extended service set, which supports roaming. Access is CSMA/CA with interframe spacing, a contention window and acknowledgements, because collisions cannot be detected on radio.
- **Bluetooth** is a related short-range ad hoc wireless LAN: piconets combine into a scatternet.